

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2024

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. (a) Expand and simplify $(2x - 4)(x + 4)$. [3]<<
>>(b) Factorise completely $9x^2 + 3x$. [2]<<
>>(c) Solve $2(x - 4) = 8$. [3]<<
>>(d) Simplify $(x^2 - 4)/(x - 2)$. [2]
[10]

2. AB is parallel to CD. Transversal PQ meets AB at X and CD at Y.<<
>>(a) Angle $AXY = 58^\circ$. Find the corresponding angle at Y. Give a reason. [2]<<
>>(b) Find the co-interior (allied) angle to 58° . Give a reason. [2]<<
>>(c) In triangle PQR, $P = 58^\circ$ and $Q = 45^\circ$. Calculate R. [2]<<
>>(d) State whether triangle PQR is acute, right-angled or obtuse. Give a reason. [2] **[8]**

3. A closed cylinder has radius 7 cm and height 8 cm. Take $\pi = 22/7$.<<
>>(a) Calculate the volume. [3]<<
>>(b) Calculate the curved surface area. [3]<<
>>(c) Calculate the total surface area. [2] **[8]**

4. The line L has equation $y = 3x + 4$.<<
>>(a) Write down the gradient of L and the coordinates of the y-intercept. [2]<<
>>(b) Find the coordinates of the point where L meets the x-axis. [2]<<
>>(c) Find the gradient of the line joining A(3, 3) and B(5, 7). [2]<<
>>(d) Write the equation of the line parallel to L that passes through (0, 1). [2]
[8]

5. The table shows values of x with frequencies f:<<
>>x=2, f=6, x=3, f=2, x=4, f=8, x=4, f=2, x=5, f=6.<<
>>(a) State the modal value of x. [1]<<
>>(b) Calculate Σf and Σfx . [3]<<
>>(c) Calculate the mean of x. [3]<<
>>(d) Find the range of x. [2] **[9]**

6. $A = \begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$ and column vector $u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
[3] (a) Calculate Au .
[2] (b) Find $\det A$.
[2] (c) Describe fully the transformation represented by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.
[2] (d) Find the image of the point $(1, 1)$ under a reflection in the y -axis. [2] **[9]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** (a) Factorise $x^2 - 5x + 4$. [3] (b) Hence solve $x^2 - 5x + 4 = 0$. [2] (c) Write down the roots of $(x - 1)(x - 4) = 0$. [1] (d) Sketch $y = (x - 1)(x - 4)$, showing the intercepts with both axes. [3] (e) State the coordinates of the turning point. [3] **[12]**
- 8.** In triangle ABC, $A = 40^\circ$, $B = 30^\circ$ and $BC = a = 8$ cm. (a) Calculate C. [2] (b) Write the sine rule. [2] (c) Calculate $AB = c$. [5] (d) State whether the triangle is acute or obtuse, with a reason. [3] **[12]**
- 9.** O is the centre of a circle. A, B and C lie on the circumference. AB is a diameter. Angle BAC = 48° . (a) Find angle ACB. Give a reason. [3] (b) Find angle ABC. [2] (c) Find angle BOC (angle at the centre standing on arc BC). Give a reason. [4] (d) Point D lies on the remaining circumference. Find angle BDC. Give a reason. [3] **[12]**
- 10.** The position vectors of A and B are $a = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $b = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$. M is the midpoint of AB. (a) Find vector AB. [3] (b) Find $|AB|$. [3] (c) Find the position vector of M. [3] (d) Show that $AM = \frac{1}{2} AB$. [3] **[12]**
- 11.** 50 candidates sat a Mathematics paper marked out of 50. A cumulative frequency curve (ogive) is drawn. (a) State the median position on the cumulative-frequency axis. [2] (b) Describe how to read the median mark from the ogive. [3] (c) The lower quartile is at position $n/4$ and the upper quartile at $3n/4$. Write these positions as numbers. [3] (d) Define the interquartile range and state one advantage of the IQR over the range. [4] **[12]**
- 12.** Triangle PQR has $P(2, 2)$, $Q(4, 2)$ and $R(2, 4)$. (a) P' is the image of P under reflection in the y-axis. Write down the coordinates of P' . [2] (b) Q is translated by the vector $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$. Find the image Q'' . [3] (c) The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ represents a transformation T. Describe T fully. [4] (d) Find $T(Q)$, the image of Q under T. [3] **[12]**
- 13.** (a) y varies directly as x . When $x = 3$, $y = 18$. (i) Find the equation connecting y and x . [3] (ii) Find y when $x = 8$. [2] (b) z varies inversely as x . When $x = 3$, $z = 12$. Let this value of z be z_0 . (i) Find z when $x = 6$. [4] (ii) Describe what happens to z when x is doubled. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).